Chintan A. Trivedi, PhD

Lab automation · scientific instrumentation · research software

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Profile

Engineer turned neuroscientist turned systems neurobiologist, who builds the hardware and software that run experiments: microscopes, closed-loop behaviour systems, automated imaging pipelines and the code that ties them to instruments and data. Senior Research Fellow at UCL; daily LLM user in development, with outputs checked against ground-truth data.

Selected projects

- **Automated genotype-to-phenotype screen.** Set up a VAST BioImager (a robot that loads larval zebrafish from well plates, orients them, images them and sorts them back into plates) and built a pipeline that keeps each larva's well and genotype identity attached from plate through registered brains to final statistics: 46 genes screened, 11 with deep phenotyping, over 1,000 3D larval brains imaged and analysed. Also designing the screen's LIMS, built with LLM coding assistance and in daily use by a 5-person team.
- **Spatial transcriptomics system (in development).** End-to-end platform: markers for HCR (in situ mRNA labelling) derived from single-cell RNA-seq, automated HCR staining in custom CAD-designed microfluidic chambers, single-molecule imaging on a Zeiss Airyscan, and a completed pipeline for cell segmentation, transcript counting and cell-type mapping.
- **pybridizer.** Open-source Python package (GitHub, PyPI) for HCR v3.0 probe design in any species, including genomes outside NCBI. Validated in a *STAR Protocols* paper establishing HCR in the malaria vector *Anopheles gambiae* (2025; ~9 citations in year one). Also used to design glia-labelling probes for a *Drosophila* glial cell atlas (*PLOS Biology*, 2023) and in zebrafish.
- **Instruments built from scratch.** Two-photon microscopes; closed-loop zebrafish virtual reality; high-speed real-time multi-larva tracking (Python/numba); Arduino-controlled multimodal behaviour rigs (visual, acoustic, thermal and chemical stimuli) and chambers of my own design.

Experience

Senior Research Fellow, Wilson Lab, University College London · Jul 2022 – present

- Co-wrote an 8-year, £4.5M Wellcome Trust grant with PIs Stephen Wilson and Isaac Bianco; promoted following the award.

Postdoctoral Research Fellow, Wilson Lab, University College London · Apr 2016 – Jun 2022

- Established the lab's behaviour work and its automated, high-throughput image analysis from scratch: designed and built behaviour rigs and imaging analysis pipelines.

Doctoral Researcher, then Postdoctoral Scholar, Max Planck Institute for Medical Research, Heidelberg · Aug 2009 – Mar 2016

- Built closed-loop VR systems for larval zebrafish, driven by muscle (EMG) recordings or by tail kinematics in partially tethered larvae, and two-photon imaging setups; performed single-cell patch clamp electrophysiology; the VR system later enabled the lab's discovery of a synaptic corollary discharge (Nature Communications, 2023); first and equal-first author papers in *Current Biology* and *Frontiers in Neural Circuits*.

Research Assistant, Texas A&M University · May 2007 – Apr 2009

Skills

- **Programming:** Python (scientific stack), LabVIEW, MATLAB
- **Lab automation and analysis:** instrument integration, LIMS design, automated imaging pipelines, microfluidics, cell segmentation, spatial and single-cell transcriptomics
- **Instrumentation and CAD:** optical and microscope design, analog/digital electronics, electrophysiology (patch clamp, EMG recordings), Arduino/microcontrollers for control and data acquisition, FreeCAD (incl. Optics Workbench)

Education

PhD (Dr. rer. nat.), summa cum laude, University of Heidelberg / Max Planck Institute for Medical Research, 2014 · **MSc Biomedical Engineering**, Texas A&M University, 2009 · **BEng Biomedical Engineering**, Saurashtra University, 2006

Publications, supervision and talks

- **Publications:** 13 peer-reviewed, h-index 11 (full list), including *STAR Protocols* (2025), *Nature Communications* (2023), *Current Biology* (2014, equal first author) and *Neuron* (2012).
- **Supervision:** around 20 BSc and MSc research students at UCL since 2016; currently on a PhD thesis committee.
- **Invited talks:** Zebrafish Neural Circuits conference, Brighton; Institute of Molecular and Cell Biology, Singapore; Institut de la Vision, Paris.